

Suresh Raghu

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EDUCATION

Indian Institute of Technology, Madras

B.Sc. in Programming & Data Science; CGPA: 3.01/4.0, Project CGPA: 4.0/4.0

Tamil Nadu, India

Oct. 2020 – Jun 2026

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science and Engineering; CGPA: 3.36/4.0

Madhya Pradesh, India

Oct. 2020 – Jun 2024

RESEARCH INTERESTS

Primary Areas: Uncertainty Quantification (UQ), Reliability & Safety of Reasoning Models, Vision-Language Models (VLMs).

Focus: Calibration & Proper Scoring for Agentic UQ, Selective Prediction & Low-Cost Uncertainty Estimation, Hallucination Detection & Visual Grounding in Reasoning VLMs.

PUBLICATIONS & PRE-PRINTS

Proper Scoring Rules for Agentic Uncertainty Quantification.

*Suresh Raghu**, *Satwik Pandey**, *Shashwat Pandey*

(Under Review)

[arXiv:2605.24756](https://arxiv.org/abs/2605.24756) | Accepted (poster) at CTB & FAGEN Workshops @ ICML 2026

- Introduced the **Trajectory Proper Score (TPS)**, a predictor-agnostic family of **strictly proper** trajectory-level scoring rules for agentic UQ, with propriety proofs under complete and administratively-censored trajectories. Showed probability recalibration shifts TPS by **~43 bootstrap SE** while moving AUROC by only **~0.3 SE** across StrategyQA, Tau2-Bench, HotpotQA, and WebShop.

SELFDUBT: Uncertainty Quantification for Reasoning LLMs via the Hedge-to-Verify Ratio.

*Satwik Pandey**, *Suresh Raghu**, *Shashwat Pandey*

(Under Review)

[arXiv:2604.06389](https://arxiv.org/abs/2604.06389) | Accepted (poster) at FAGEN Workshop @ ICML 2026

- Proposed the **Hedge-to-Verify Ratio (HVR)**, a single-pass $\mathcal{O}(1)$ uncertainty signal that scores a reasoning trace by how much it hedges versus self-verifies. Traces with no hedging language are correct **96.1% of the time** (at 25.4% coverage), and fusing HVR with the model's verbalized confidence beats sampling-based Semantic Entropy on discrimination ($p = 0.001$) at **10× lower inference cost**, across 7 models and 3 benchmarks.

Don't Blink: Evidence Collapse during Multimodal Reasoning.

*Suresh Raghu**, *Satwik Pandey**

(Under Review)

[arXiv:2604.04207](https://arxiv.org/abs/2604.04207) | * Equal contribution

- Identified “**evidence collapse**”, a universal (9/9 model×dataset cells) decay of visual grounding during VLM reasoning that text-only entropy cannot detect, and designed a task-conditional **vision veto** that cuts selective risk by up to **1.9 pp** at 90% coverage on MathVista, HallusionBench, and MMMU_Pro.

RESEARCH & ENGINEERING EXPERIENCE

VFS Global

Manager – AI (Founding Lead, AI Research Engineering)

New Delhi, India

May 2024 – Present

- **Adaptive Query Routing:** Architected the document-extraction pipeline around an adaptive router that sends deterministic documents to lightweight parsers and ambiguous ones to reasoning VLMs, reaching **99.1% field-level extraction accuracy**.
- **Uncertainty-Gated Extraction Cascade:** Used a cheap token-logit trigger to run **Semantic Entropy** and Monte Carlo self-consistency only on suspect outputs, escalating flagged cases to a larger LLM and cutting critical extraction errors by **87%**.
- **Biometric Compliance Pipeline:** Proposed and built a two-stage visa photo-verification system consisting of a cheap image-quality gate (Laplacian blur detection) followed by a VLM stage for pose/occlusion checks, reducing cases requiring manual intervention by **78%**.

Adani Digital Labs

Applied Machine Learning Intern

Gurugram, India

June 2023 – April 2024

- **Self-Supervised Representation Learning:** Used Meta's **DINOv2** embeddings with an SVM over latent-space projections for few-shot image clustering, replacing manual tagging at **93% precision**.
- **Multimodal Search:** Built a CLIP + GPT-4 Vision search engine supporting both text and image queries over **proprietary retail inventory**.